INCH-POUND

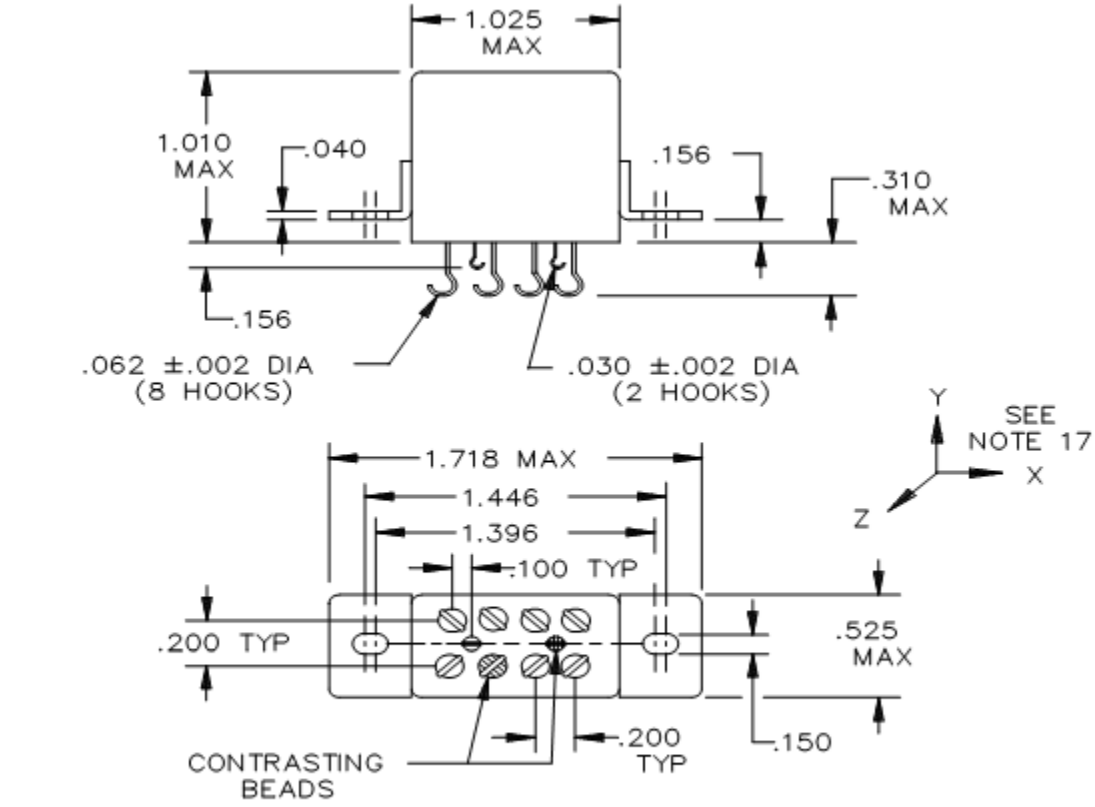
MIL-PRF-83536/12F
11 October 2023
SUPERSEDING
MIL-PRF-83536/12E
26 March 2020

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT, LOW LEVEL TO 10 AMPERES, MAGNETIC LATCH, HERMETICALLY SEALED, ALL WELDED, DC COILS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the latest issue of MIL-PRF-83536.



RAISED VERTICAL FLANGE MOUNT (RVFM) WITH SOLDER HOOKS

FIGURE 1. Dimensions and configurations.

AMSC N/A
FSC 5945

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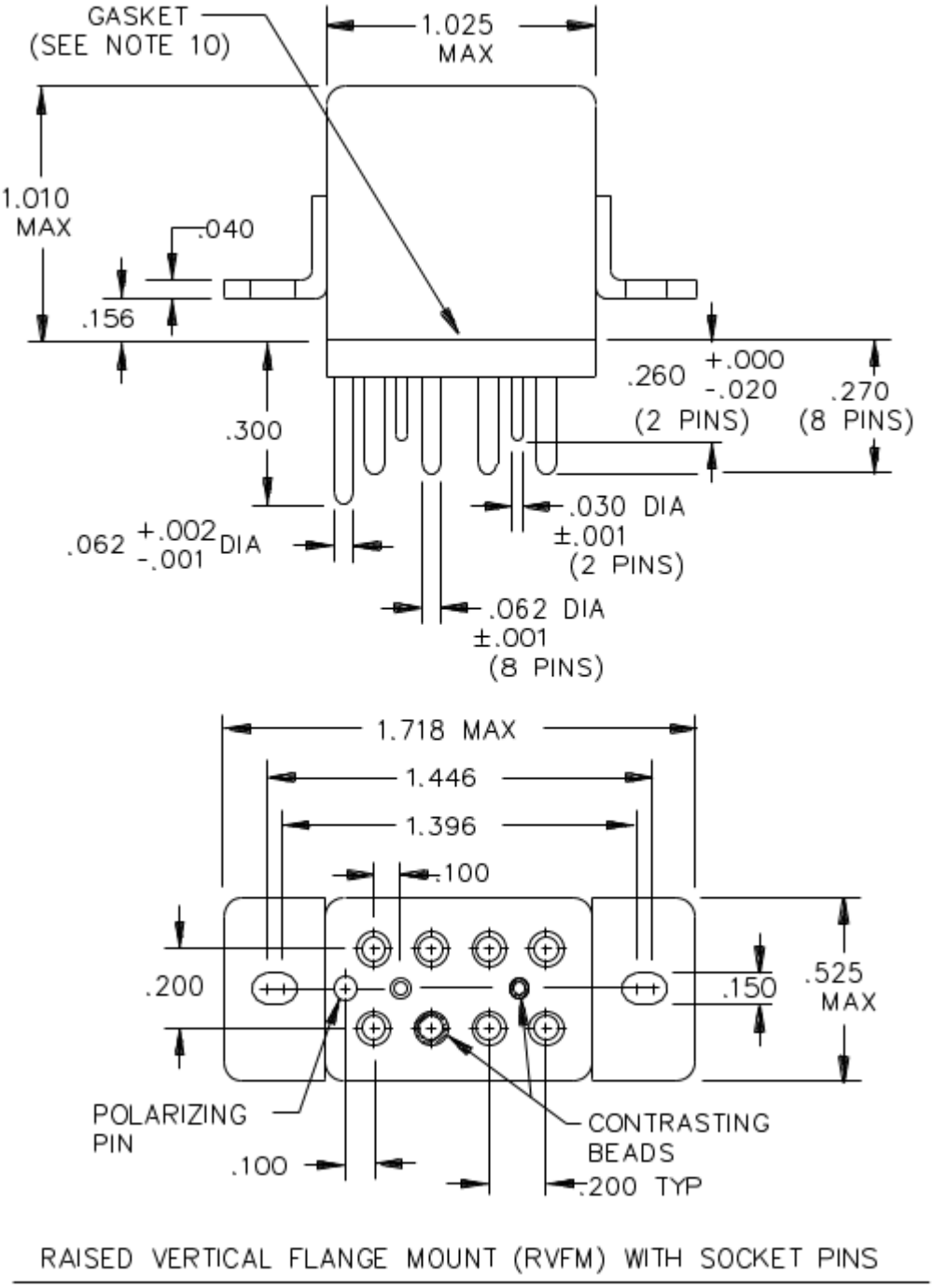


FIGURE 1. Dimensions and configurations - Continued.

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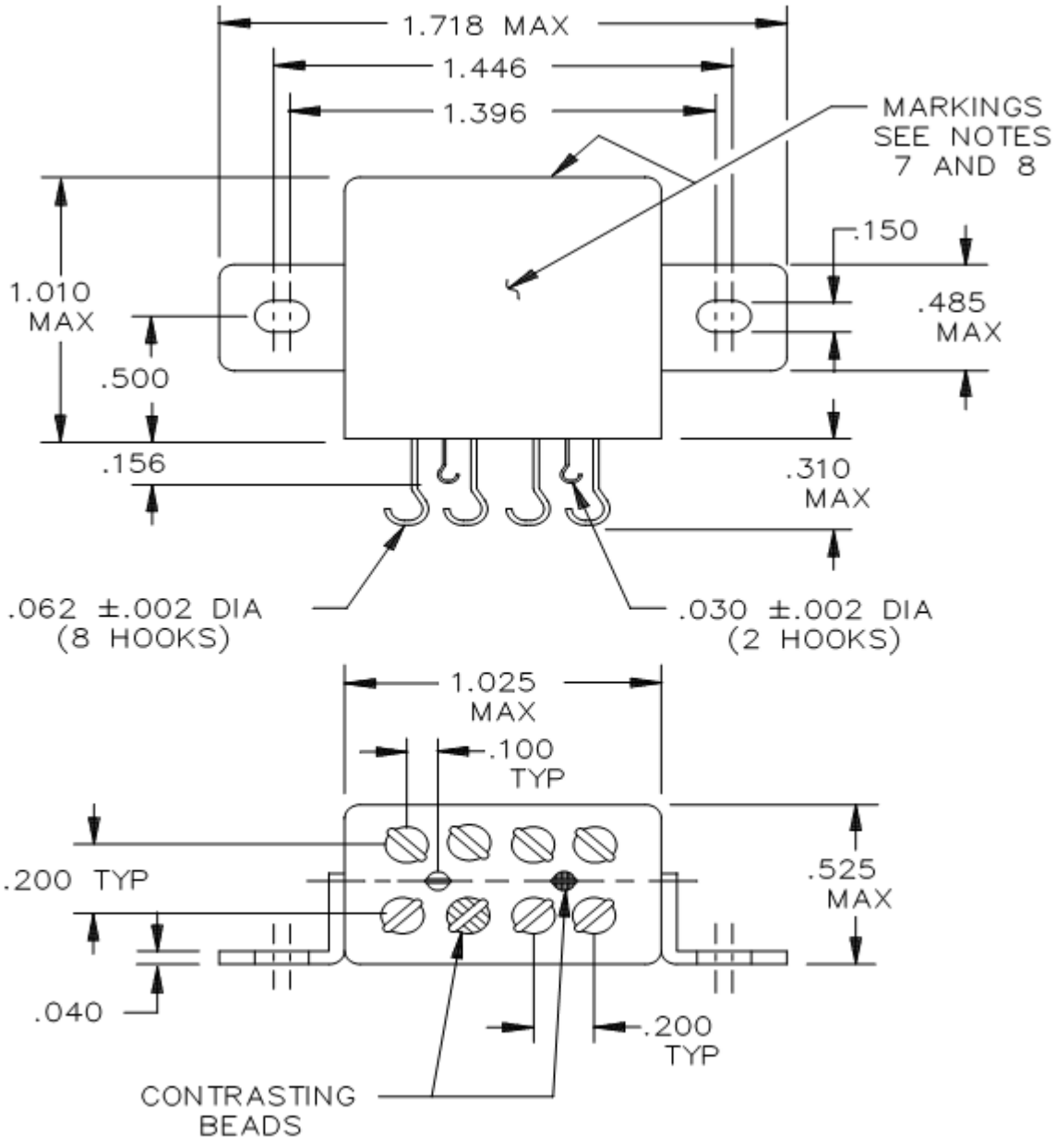
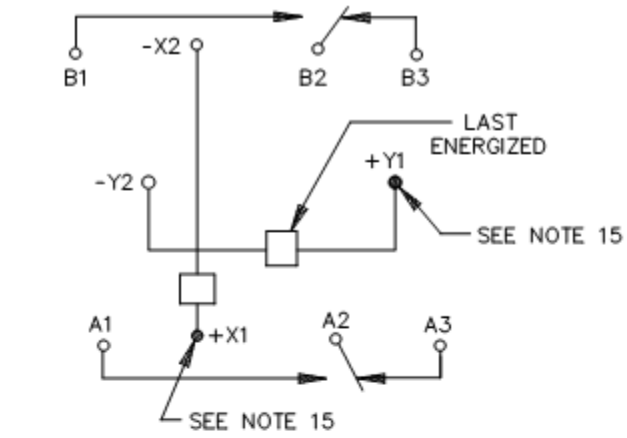
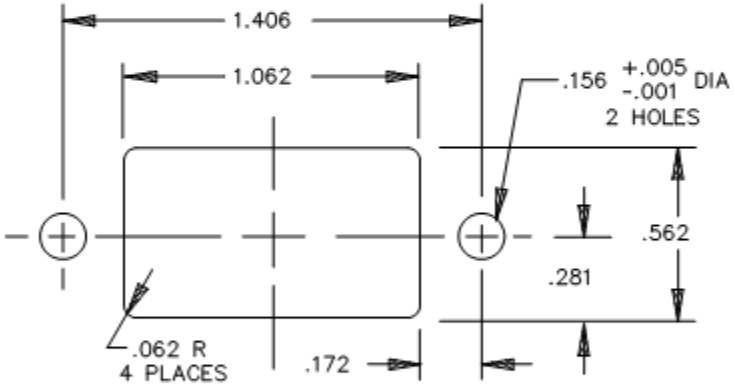


FIGURE 1. Dimensions and configurations - Continued.

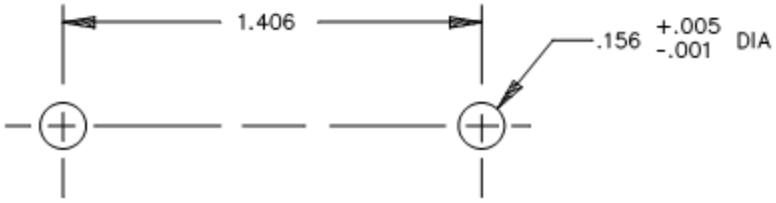
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CIRCUIT DIAGRAM (RVFM AND HFM)



MOUNTING HOLE LAYOUT (RVFM)



MOUNTING HOLE LAYOUT (HFM)

FIGURE 1. Dimensions and configurations - Continued.

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Inches	mm	Inches	mm	Inches	mm
0.000	0.00	0.156	3.96	0.500	12.70
0.001	0.03	0.172	4.37	0.525	13.34
0.002	0.05	0.200	5.08	0.562	14.27
0.005	0.13	0.260	6.60	1.025	26.04
0.020	0.51	0.270	6.86	1.062	26.98
0.030	0.76	0.281	7.14	1.396	35.46
0.040	1.02	0.300	7.62	1.406	35.71
0.062	1.57	0.310	7.87	1.446	36.73
0.100	2.54	0.312	7.92	1.718	43.64
0.150	3.81	0.485	12.32		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± 0.010 (0.25 mm).
4. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
5. These relays are polarized bistable.
6. This relay shall not be damaged by reverse polarity.
7. Applicable to horizontal flange mount only. The circuit diagram, manufacturer's PIN, and the military PIN shall be marked on the near side. The remaining portion of the nameplate date shall be marked on the far side.
8. Applicable to horizontal flange mount only. Relays shall be marked with the manufacturer's name or CAGE code and date code. Marking shall be with the bottom of the print adjacent to the near side.
9. Socket pin terminals shall provide the operational, environmental, and interface characteristics to provide a reliable interconnect to gold-plated contacts. Terminals, except the polarizing pin, shall be gold plated. One system for gold plating that may be used is ASTM-B488, type 3, class 1.25 with a nickel underplate of 50 to 150 microinches thick. The gold plating system shall enable the product to meet the performance requirements of this specification and shall be approved by the qualifying activity.
10. Gasket shall provide a reliable seal between the relay and mating socket that will meet the environmental, operational, and interface requirements of the relay with the mating socket. The gasket shall have shore hardness 15 to 35, thickness 0.050 ± 0.010 (1.27 ± 0.25 mm). Gasket material according to SAE-AMS3332 has been considered acceptable.
11. Relay is magnetically latched in both positions.
12. All hooks shall be tin finished.
13. Relays shall have a (+) sign placed on the circuit diagram as shown.
14. Coil symbol optional in accordance with MIL-STD-1285.
15. Indicated terminals shall be identified by a contrasting bead.
16. Direction of hook openings is optional.
17. Plane of critical motion for vibration and shock is Y-axis.

FIGURE 1. Dimensions and configurations – Continued

REQUIREMENTS:

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Contact data:

Load ratings: See table I.

Low level: 10 to 50 μA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Mixed loads: Applicable.

TABLE I. Rated contact load and cycles (amperes per pole).

Type of load (high level)	Cycles x 10^3	28 V dc	115 V ac 1 phase 400 Hz	115 V ac 1 phase 50/60 Hz 1/	115/200 V ac 3 phase 400 Hz	115/200 V ac 3 phase 50/60 Hz 1/
Resistive	100	10	10	2.5	10	2.5
Inductive	20	8	8	N/A	8	N/A
Inductive	10	N/A	N/A	2.5	N/A	2.5
Motor	100	4	4	2	4	2
Lamp	100	2	2	1	N/A	N/A

1/ For 50/60 Hz rating, rupture and overload are not applicable and life for each load shall be 10,000 cycles.

Life: 100,000 cycles, unless otherwise specified (see table I).

Contact voltage drop and resistance:

Initial contact voltage drop: 0.100 volt maximum. Rated resistive current at 6 V dc or peak ac. Relays shall not make or break this load.

Initial contact resistance: 0.010 ohm maximum. 50 mA max at 6 V dc or peak ac.

High level life (contact voltage drop):

During life: 10 percent open circuit voltage maximum.

After life: 0.125 volt maximum.

Intermediate current (contact resistance):

During intermediate current: 3 ohms maximum.

After intermediate current: 0.150 ohm maximum.

Low level (contact resistance):

During low level: 100 ohms maximum.

After low level: 0.150 ohm maximum at 100 mA and 28 V dc.

Contact bounce: 1.0 millisecond maximum.

Contact stabilization time: 2.5 milliseconds maximum.

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Overload current: 40 amperes dc (60 amperes ac).

Rupture current: 50 amperes dc (80 amperes ac).

Time current relay characteristics: See table II.

TABLE II. Time current relay characteristics. 1/

Successive application	Amperes	Time
1	15	1 hour
2	50	5.0 seconds
3	100	1.2 seconds
4	250	0.2 second
5	350	0.1 second

1/ All relays shall withstand overload and fault currents. Relays must be able to sustain five applications (make and carry only) of power concurrently on adjacent poles at each of five different current levels in the sequence listed in table II. Separate relays shall be tested at 28 V dc and 115/200 V ac, 400 Hz 3-phase. Cooling time between successive applications shall be 30 minutes. Tests shall be performed on both normally open and normally closed contacts. There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at the conclusion.

Coil data 1/: See table III.

Operate time: 10 milliseconds maximum (each coil) with rated coil voltage over the temperature range.

Duty rating: Continuous.

Coil transient suppression: Not applicable.

Neutral screen: Applicable.

Electrical data:

Insulation resistance:

Initial: 100 megohms minimum at 500 V dc.

After life or environmental tests: 50 megohms minimum at 500 V dc.

1/ CAUTION: Due to possible interaction of relay magnetic fields, the following spacing requirements, as a minimum, shall be considered in dense packaging situations:

a. Row to row assisting fields: 0.125 (3.18 mm).

b. Row to row opposing fields: 0.1875 (4.763 mm).

c. Side to side alternating fields: 0.0625 (1.588 mm).

d. Side to side like fields: 0.125 (3.18 mm).

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TABLE III. Dash numbers and characteristics. 1/

Dash number 2/		Mount	Coil data						
			At 25°C				Over temperature range	Maximum latch/reset voltage	
Solder hook	Socket pin		Coil voltage (V dc) 3/		Coil resistance (ohms) minimum	Specified latch/reset voltage (V dc) 4/	Specified latch/reset voltage (V dc) 4/	High temperature test	Continuous current test
			Rated	Max					
001	002	RVFM	6	7.3	34	3.2	4.5	5.0	5.7
003	---	HFM	6	7.3	34	3.2	4.5	5.0	5.7
004	005	RVFM	12	14.5	135	6.5	9.0	9.9	11.25
006	---	HFM	12	14.5	135	6.5	9.0	9.9	11.25
007	008	RVFM	28	29 5/	400	14.5	18.0	19.8	22.5
009	---	HFM	28	29 5/	400	14.5	18.0	19.8	22.5
010	011	RVFM	48	50	1,400	24.0	36.0	38.0	42.0
012	---	HFM	48	50	1,400	24.0	36.0	38.0	42.0

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications.
2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be L, 3.0; M, 1.0; P, 0.1; R, 0.01. Example: 001L, 002R.
3/ CAUTION: The use of any coil voltages less than the rated coil voltage will compromise the operation of the relay.
4/ Pickup voltage as shown is for test purposes only and is not to be used for design criteria.
5/ When maximum ambient temperature does not exceed +85°C, the maximum coil voltage shall be 32 V dc.

Dielectric withstanding voltage 3/:

	All others	
	Coil to coil Coil to case	All other points
Sea level:		
Initial:	1,000	1,250
After life:	1,000	1,000
Altitude:		
At 80,000 feet:	250	350
At 300,000 feet:	500	500

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3/ Dielectric may be improved by suitable insulation of terminals and wiring after installation.

Environmental characteristics:

Temperature range: -70°C to +125°C.

Maximum altitude rating: 300,000 feet.

Shock (specified pulse): Applicable, MIL-STD-202-213, test condition C, except peak value shall be 200 g's for 6 ±1 ms. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Vibration (sinusoidal): Applicable, MIL-STD-202-204, 30 g's, except frequency range shall be 10 to 3,000 Hz. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Vibration (random): Applicable to qualification and group C inspection. Test in accordance with MIL-STD-202-214, test condition IG (0.4 g^2/Hz, 50 to 2,000 Hz). Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Acceleration: Applicable, except 15 g's.

Physical data:

Dimensions and configurations: See figure 1.

Weight: .090 pound (40.9 grams).

Terminal strength:

Solder hook terminals:

Pull force:

For 0.030 diameter terminals: 3 pounds ±0.3 pound.

For 0.062 diameter terminals: 10 pounds ±1.0 pound.

Bend: Applicable.

Twist: Applicable to coil terminals Y1 and Y2 only.

Solder pin terminals:

Pull force:

For 0.031 diameter terminals: 3 pounds ±0.3 pound (applicable to track mount).

For 0.062 diameter terminals: 10 pounds ±1.0 pound.

Bend: Applicable.

Twist: Applicable to coil terminals Y1 and Y2 only.

Socket pin terminals:

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Pull force:

For 0.031 diameter terminals: 3 pounds ±0.3 pound (applicable to track mount).

For 0.062 diameter terminals: 10 pounds ±1.0 pound.

Bend: Applicable.

Twist: Applicable to coil terminals Y1 and Y2 only.

Terminal solderability: Applicable to solder hook terminals only.

Seal: Hermetic.

Marking: Applicable.

Part or Identifying Number (PIN): M83536/12 (dash number from table III and suffix letter designating failure rate level).

Qualification inspection:

Qualification inspection and sample size: See table IV.

TABLE IV. Qualification inspection and sample size. 1/

Single submission	Group submission	
18 units plus 1 open unit for level L at C = 0 2/	M83536/12-008	18 units plus 1 open unit for level L at C = 0 2/
33 units plus 1 open unit for level M at C = 0 2/	M83536/12-008	33 units plus 1 open unit for level M at C = 0 2/
Qualification inspection as applicable	M83536/12-008	Qualification inspection as applicable
	M83536/12-009	2 units, qualification inspection table, group II, shock, vibration, acceleration, terminal strength, and seal.
	M83536/12-002 M83536/12-006 M83536/12-012	2 units, qualification inspection table, group II.

1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-PRF-83536/13 may be used in addition to MIL-PRF-83536/12 data. Prior to performance of retention of qualification testing, the relay manufacturer shall preselect the sampling plan.

2/ The number of units required for qualification testing shall be increased as required in group V, table II of MIL-PRF-83536, if the contractor elects to test the number of units permitting one or more failures. Prior to performance of qualification testing, the relay manufacturer shall preselect the sampling plan.

Qualification inspection (reduced testing) (sample size: 2 units each coil voltage and 1 unsealed unit). See table V.

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If the relays produced for MIL-PRF-83536/12 are similar in construction and design, except for the supersession network to the relays produced for MIL-PRF-83536/13, then reduced testing for qualification of MIL-PRF-83536/12 relays may be performed concurrent with or subsequent to successful qualification of MIL-PRF-83536/13 relays. For reduced testing, see table V.

TABLE V. Qualification inspection (reduced testing).

Inspection
2 units each coil voltage Group II of qualification inspection table 1 additional unsealed sample unit for internal inspection

Supersession data: See table VI.

TABLE VI. Supersession data.

Superseded PIN	Replacement PIN M83536/12-
M6106/50-001	007
002	008
003	009

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Referenced documents. In addition to MIL-PRF-83536, this document references the following:

ASTM B488
SAE-AMS3332
MIL-STD-202-213
MIL-STD-1285
MIL-STD-202-204
MIL-STD-202-214
MIL-PRF-83536/13

Custodians:
Army - CR
Navy - EC
Air Force -85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2023-061)

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